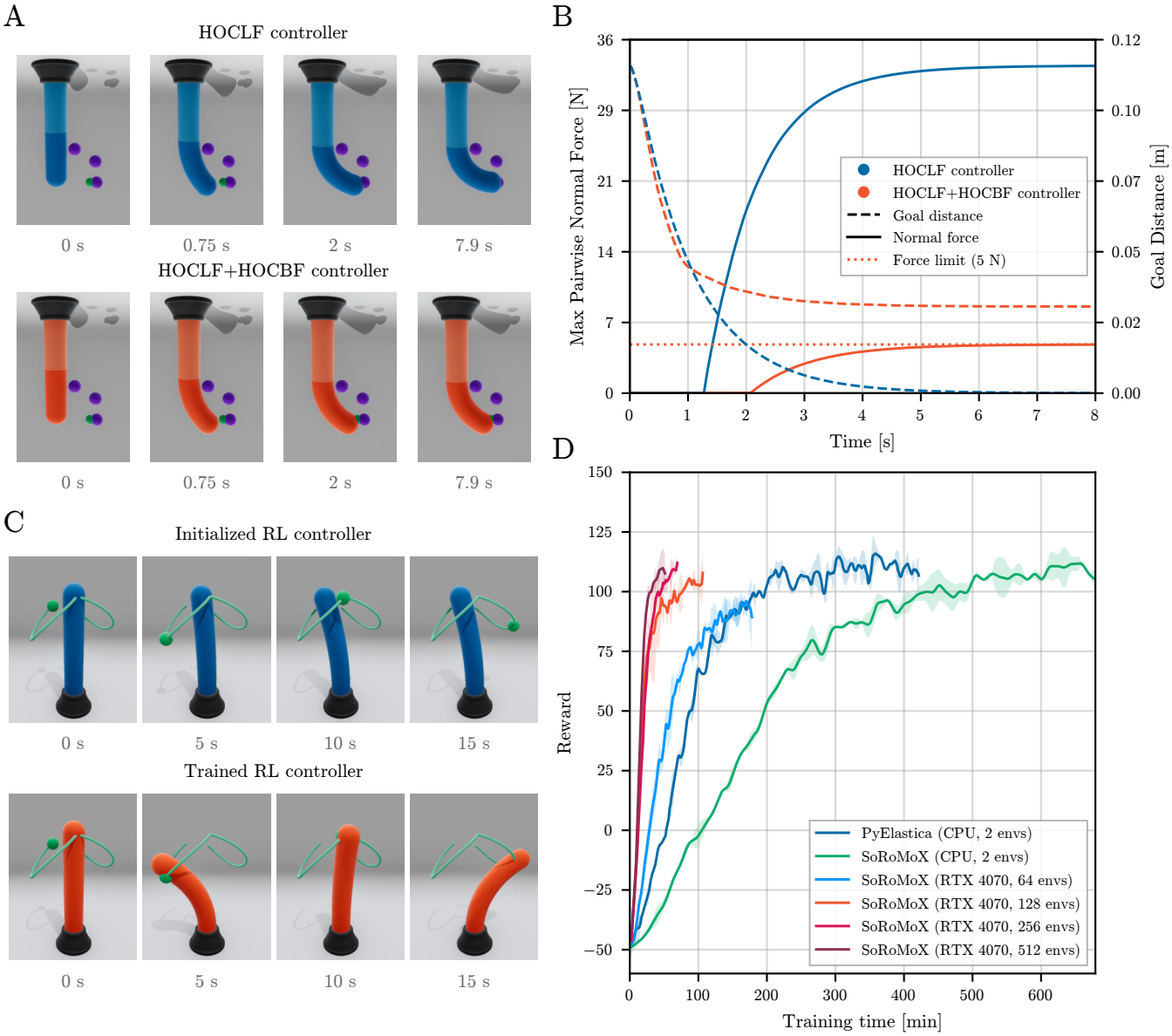

A
HOCLF controller
0 s
0.75 s
2 s
7.9 s
HOCLF+HOCBF controller
0 s
0.75 s
2 s
7.9 s
B
Max Pairwise Normal Force [N]
Goal Distance [m]
Time [s]
HOCLF controller
HOCLF+HOCBF controller
Goal distance
Normal force
Force limit (5 N)
C
Initialized RL controller
0 s
5 s
10 s
15 s
Trained RL controller
0 s
5 s
10 s
15 s
D
Reward
Training time [min]
PyElastica (CPU, 2 envs)
SoRoMoX (CPU, 2 envs)
SoRoMoX (RTX 4070, 64 envs)
SoRoMoX (RTX 4070, 128 envs)
SoRoMoX (RTX 4070, 256 envs)
SoRoMoX (RTX 4070, 512 envs)